



A stratified analysis of a deep learning algorithm in the diagnosis of diabetic retinopathy in a real-world study

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Abstract

Background: The aim of our research was to prospectively explore the clinical value of a deep learning algorithm (DLA) to detect referable diabetic retinopathy (DR) in different subgroups stratified by types of diabetes, blood pressure, sex, BMI, age, glycosylated hemoglobin (HbA1c), diabetes duration, urine albumin-to-creatinine ratio (UACR), and estimated glomerular filtration rate (eGFR) at a real-world diabetes center in China.

Methods: A total of 1147 diabetic patients from Shanghai General Hospital were recruited from October 2018 to August 2019. Retinal fundus images were graded by the DLA, and the detection of referable DR (moderate non-proliferative DR or worse) was compared with a reference standard generated by one certified retinal specialist with more than 12 years of experience. The performance of DLA across different subgroups stratified by types of diabetes, blood pressure, sex, BMI, age, HbA1c, diabetes duration, UACR, and eGFR was evaluated.

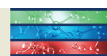
Results: For all 1674 gradable images, the area under the receiver operating curve, sensitivity, and specificity of the DLA for referable DR were 0.942 (95% CI, 0.920-0.964), 85.1% (95% CI, 83.4%-86.8%), and 95.6% (95% CI, 94.6%-96.6%), respectively. The DLA showed consistent performance across most subgroups, while it showed superior performance in the subgroups of patients with type 1 diabetes, UACR ≥ 30 mg/g, and eGFR < 90 mL/min/1.73m².

Conclusions: This study showed that the DLA was a reliable alternative method for the detection of referable DR and performed superior in patients with type 1 diabetes and diabetic nephropathy who were prone to DR.

Na Li, Mingming Ma, and Mengyu Lai contributed equally to this work and are considered to be co-first authors.

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KEYWORDS

deep learning algorithm, diabetic retinopathy, referable DR, retinal fundus images

Highlights

- Our deep learning algorithm (DLA)-based diabetic retinopathy (DR) grading software showed excellent performance in detecting referable DR, especially in patients with type 1 diabetes and diabetic nephropathy who were prone to DR.
- The DLA showed consistent performance in the detection of referable DR across different subgroups stratified by blood pressure, sex, BMI, age, glycosylated hemoglobin, and diabetes duration in a real-world diabetic center.

1 | INTRODUCTION

The prevalence of diabetes mellitus (DM) is increasing worldwide, and it has been reported to be 12.8% in Chinese adults.¹ Diabetic retinopathy (DR), which is a common but serious complication of diabetes, is the leading cause of blindness worldwide.^{2,3} The prevalence of DR among DM patients has been estimated at 34.6%,² and the number of DR patients is growing.⁴ Sight-threatening DR can be avoided when detected early through screening strategies by regular clinical examination or grading of retinal photographs and treated in a timely fashion.^{5,6}

There are different methods of DR screening in the world, including direct and indirect ophthalmoscopy, digital fundus photography, fundus fluorescein angiography, and other examinations.^{7,8} Recently, deep learning algorithms (DLA), a branch of artificial intelligence (AI), has been widely applied in image recognition, speech recognition, and natural language processing.^{9,10} For DR detection, DLA have demonstrated excellent sensitivity and specificity, and have been shown to produce expert-level diagnoses for grading fundus photographs.^{4,11–16} However, a majority of DLA systems have been validated using online curated or publicly available datasets (EyePACS, Messidor-2, e-optha),^{11,12,15} which contained high-quality photographs from individuals.

DR was strongly associated with chronic hyperglycemia, diabetic duration, hypertension, and nephropathy.¹⁷ It was reported that the incidence of DR was higher in patients with type 1 diabetes than in those with type 2 diabetes.¹⁸ Retinopathy caused by hypertension may interfere with the diagnosis of DR. However, few studies have been conducted on the performance of DLA in different settings. To our knowledge, only Ting et al have reported that their DLA showed comparable performance in different subgroups of patients stratified by age, sex, and glycosylated hemoglobin (HbA1c).¹³

Therefore, in this study, we conducted a prospective clinic-based DR screening in the real world, using automated DR grading software (an AI-based DLA) to grade more than 2000 retinal photographs of patients with diabetes collected by Shanghai General Hospital. The diagnostic accuracy of the DLA was validated, and the performance of the DLA across different subgroups stratified by types of diabetes, blood pressure (BP), sex, BMI, age, HbA1c, diabetes duration, urine albumin-to-creatinine ratio (UACR), and estimated glomerular filtration rate (eGFR) was evaluated. We also analyzed the reasons for ungradable images and the inconsistency between the DLA and the retinal specialist.

2 | METHODS

2.1 | Study design, population, and imaging

This study recruited patients with diabetes from the Department of Endocrinology of Shanghai General Hospital between October 2018 and August 2019. Patients who were pregnant at the time or had any history of intraocular surgery other than cataract surgery in the past year were excluded from the study. Ultimately, a total of 1147 patients (2286 eyes) were enrolled.

Clinical data including age, sex, weight, height, BMI, BP, diabetes duration, total cholesterol (TC), high-density lipoprotein (HDL), low-density lipoprotein (LDL), triglycerides (TG), HbA1c, serum creatinine, urine albumin, and urine creatinine were recorded. UACR was calculated using the following formula: UACR = urine albumin/urine creatinine. The eGFR was calculated using the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation.¹⁹



For each patient, macula-centered 45° color fundus photographs of each eye were taken using a retinal fundus camera (KOWA nonmyd WX, Tokyo, Japan). No mydriatic agents were applied. This study was approved by the hospital ethics committee, and all participants signed written informed consent.

2.2 | Automated DR grading software

The automated DR grading software used in this study (VoxelCloud, China) was developed using deep learning techniques. Two different networks were included in the software: DR classification network and quality control network. The DR classification network, which was the crucial component of the automated DR grading software, was trained on two datasets. The first dataset (Eyepacs dataset) came from an extensive private retinal image database obtained between 2005 and 2015, containing 140 000 fundus photographs of approximately 37 000 patients, which was used to train the initial DR grading model. The images were assigned retinopathy severity levels based on the International Clinical Diabetic Retinopathy Severity (ICDRS) scale,²⁰ which was developed by the International Council of Ophthalmology and adopted by the American Academy of Ophthalmology.²¹ The second dataset (domestic fundus dataset), obtained from a public hospital in China (not from Shanghai General Hospital, different from the dataset obtained between October 2018 and August 2019), contained approximately 1200 color fundus images, and the DR severity grade was assigned based on the consensus among three retinal specialists. These data were selected to improve the performance of our model in complex situations.

The quality control network was trained on 6400 fundus photographs with different image quality, which was a subset of the first dataset used for training the DR classification network. Architecture and training details of both networks are shown in Supplementary Methods.

All color fundus images were resized to a standard resolution of 800 by 800 pixels and normalized to pixel intensity values between 0 and 1 before being processed by the software.

2.3 | Reference standard grading

The reference standard for DR was generated by one certified retinal specialist with more than 12 years of experience; this specialist assigned the grades based on the ICDRS scale,²⁰ which uses a 5-point grading system: no DR, mild nonproliferative DR (NPDR), moderate NPDR,

severe NPDR, and proliferative DR (PDR). The retinal specialist was blinded to the results of the automated DR grading software. Referable DR was defined as moderate NPDR or worse.

All images were graded by both the automated DR grading software and the retinal specialist. Then, the performance of the DLA-based DR grading software was compared to the reference standards. To compare the diagnostic performance of the DLA in different subgroups, we categorized the patients into different subgroups according to type of diabetes (type 1 diabetes, type 2 diabetes), history of hypertension (presence or absence of high blood pressure [HBP]), sex (female, male), BMI ($<24 \text{ kg/m}^2$, $\geq 24 \text{ kg/m}^2$), age (≤ 40 years, >40 years and ≤ 60 years, >60 years), HbA1c ($<7\%$ [53 mmol/mol], $\geq 7\%$ [53 mmol/mol] and $<9\%$ [75 mmol/mol], $\geq 9\%$ [75 mmol/mol]), diabetes duration (<1 year, ≥ 1 year, ≥ 5 years, ≥ 10 years), UACR ($<30 \text{ mg/g}$, $\geq 30 \text{ mg/g}$), and eGFR ($\geq 90 \text{ mL/min/1.73m}^2$, $<90 \text{ mL/min/1.73m}^2$).

2.4 | Statistical analysis

Variables were expressed as the mean and SD or as the median and interquartile range (25%-75%) as appropriate. One-way analysis of variance or the Kruskal-Wallis test was used to compare differences in continuous variables among DR stages (no DR, mild NPDR, moderate NPDR, severe NPDR, and PDR). The Mann-Kendall test was employed to investigate the trends between DR stages and patients' demographic and clinical characteristics. The sensitivity, specificity, and area under the receiver operating curve (AUC) with 95% CI of the DLA in detecting referable DR were calculated and compared to the reference standard at the level of individual eyes. Analyses were performed in R V.3.6.0 (R Foundation for Statistical Computing, Vienna, Austria).

3 | RESULTS

The overall demographics and clinical characteristics of the patients are listed in Table 1. In total, 1147 patients with diabetes were enrolled in this study, including 36 with type 1 diabetes and 1111 with type 2 diabetes. The mean age of the patients was 50 ± 12 years, and 68.4% of the patients were male. The duration of diabetes in the study population was 2.08 (0.08-9.12) years, HbA1c was $8.26 \pm 2.07\%$ ($66.8 \pm 22.6 \text{ mmol/mol}$), BMI was $25.67 \pm 3.58 \text{ kg/m}^2$, systolic blood pressure (SBP) was $128 \pm 16 \text{ mm Hg}$, and diastolic blood pressure (DBP) was $78 \pm 10 \text{ mm Hg}$. Of the 1147 patients, 772 (67.3%) had no DR, 143 (12.5%) had mild NPDR, 93 (8.1%) had moderate

TABLE 1 Demographic and clinical characteristics of patients with diabetes stratified by different DR stages assigned by the retinal specialist

		Non-referable DR		Referable DR				ANOVA/ Kruskal-Wallis (<i>P</i> value)	Kendall's tau-b
		No DR	Mild NPDR	Moderate NPDR	Severe NPDR	PDR	Ungradable		
No. of patients (%)	1147 (100%)	772 (67.3%)	143 (12.5%)	93 (8.1%)	43 (3.7%)	20 (1.7%)	76 (6.6%)	-	-
No. of type 1 diabetes (%)	36	20 (55.5%)	4 (11.1%)	1 (2.8%)	1 (2.8%)	5 (13.9%)	5 (13.9%)	-	-
No. of type 2 diabetes (%)	1111	752 (67.6%)	139 (12.5%)	92 (8.3%)	42 (3.8%)	15 (1.4%)	71 (6.4%)	-	-
Age, mean (SD), y	50 ± 12	49 ± 12	50 ± 11	52 ± 12	56 ± 10	49 ± 9	59 ± 12	<0.001	0.07, <i>P</i> < .001
Men, n (%)	784 (68.4%)	523	103	70	25	14	49	-	-
BMI (kg/m ²)	25.67 ± 3.58	25.69 ± 3.60	25.87 ± 3.50	25.20 ± 3.84	25.33 ± 3.14	24.80 ± 3.05	26.00 ± 3.62	0.46	-0.02, <i>P</i> = .38
Diabetes duration, median (IQR), y	2.08 (0.08-9.12)	1.17 (0.00-6.10)	3.00 (0.25-10.17)	8.17 (1.00-13.42)	10.17 (2.21-15.75)	9.79 (0.42-14.42)	7.17 (1.35-15.17)	<0.001	0.19, <i>P</i> < .001
SBP (mm Hg)	128 ± 16	127 ± 15	130 ± 18	131 ± 17	134 ± 19	133 ± 20	133 ± 18	<0.001	0.08, <i>P</i> < .001
DBP (mm Hg)	78 ± 10	77 ± 10	80 ± 11	78 ± 10	79 ± 12	78 ± 12	75 ± 10	0.11	0.05, <i>P</i> = .04
HbA1c (%)	8.26 ± 2.07	8.18 ± 2.12	8.48 ± 2.08	8.46 ± 1.80	9.09 ± 2.03	8.46 ± 1.93	7.91 ± 1.75	0.03	0.09, <i>P</i> < .001
HbA1c (mmol/mol)	66.8 ± 22.6	65.9 ± 23.2	69.2 ± 22.7	69.0 ± 19.7	75.8 ± 22.2	69.0 ± 21.1	62.9 ± 19.1	0.03	0.09, <i>P</i> < .001
TC (mmol/L)	4.83 ± 1.29	4.91 ± 1.26	4.70 ± 1.10	4.67 ± 1.22	4.65 ± 1.23	4.47 ± 1.40	4.67 ± 1.89	0.08	-0.07, <i>P</i> < .001
HDL (mmol/L)	1.01 ± 0.28	1.01 ± 0.26	1.00 ± 0.29	1.04 ± 0.35	1.03 ± 0.32	1.03 ± 0.39	1.04 ± 0.32	0.87	-0.02, <i>P</i> = .36
LDL (mmol/L)	2.68 ± 0.92	2.75 ± 0.91	2.66 ± 0.93	2.53 ± 1.00	2.47 ± 0.75	2.40 ± 1.16	2.38 ± 0.86	0.03	-0.08, <i>P</i> < .001
TG (mmol/L)	2.15 ± 2.82	2.18 ± 2.71	2.08 ± 2.77	2.02 ± 1.76	1.88 ± 1.73	1.51 ± 0.88	2.41 ± 5.15	0.73	-0.04, <i>P</i> = .07
UACR (mg/g)	71.4 ± 327.6	44.8 ± 167.5	54.5 ± 161.1	75.0 ± 207.6	216.9 ± 484.6	439.1 ± 1455.6	186.1 ± 702.4	<0.001	0.15, <i>P</i> < .001
eGFR (mL/min/1.73m ²)	113.0 ± 19.2	115.3 ± 17.0	114.2 ± 19.3	108.7 ± 21.9	99.0 ± 23.8	108.8 ± 26.3	100.5 ± 23.4	<0.001	-0.09, <i>P</i> < .001

Note: Data are presented as the mean ± SD or median (IQR) as appropriate. Ungradable: insufficient image quality.

Abbreviations: ANOVA, analysis of variance; DBP, diastolic blood pressure; DR, diabetic retinopathy; eGFR, estimated glomerular filtration rate; HbA1c, glycosylated hemoglobin; HDL, high-density lipoprotein; IQR, interquartile range; LDL, low-density lipoprotein; NPDR, nonproliferative DR; PDR, proliferative DR; SBP, systolic blood pressure; TC, total cholesterol; TG, triglycerides; UACR, urine albumin-to-creatinine ratio.



NPDR, 43 (3.7%) had severe NPDR, 20 (1.7%) had PDR, and 76 (6.6%) were considered ungradable because both eyes had insufficient fundus image quality graded by the retinal specialist per the ICDR grading system. Among the gradable patients, the prevalence of any DR and referable DR were 27.9% and 14.6%, respectively. Age ($P < .001$), duration of diabetes ($P < .001$), SBP ($P < .001$), HbA1c ($P = .03$), LDL ($P = .03$), UACR ($P < .001$), and eGFR ($P < .001$) were significantly different among the DR stages, while there were no significant differences in BMI, DBP, TC, HDL, or TG among the DR stages ($P > .05$) (Table 1). The correlations of DR stages with age (Kendall's tau-b = 0.07, $P < .001$), duration of diabetes (Kendall's tau-b = 0.19, $P < .001$), SBP (Kendall's tau-b = 0.08, $P < .001$), DBP (Kendall's tau-b = 0.05, $P = .04$), HbA1c (Kendall's tau-b = 0.09, $P < .001$) and UACR (Kendall's tau-b = 0.15, $P < .001$) were significantly positive, while the correlations of DR stages with TC (Kendall's tau-b = -0.07 , $P < .001$), LDL (Kendall's tau-b = -0.08 , $P < .001$), and eGFR (Kendall's tau-b = -0.09 , $P < .001$) were negative (Table 1).

The final analysis included 2286 images from 1147 patients. Eight patients had only one eye. The distributions of the grades assigned by the retinal specialist and the DLA were compared, and the weighted kappa score was 0.771 (95% CI, 0.7288-0.8131). Of the 2286 images, 223 (9.8%) could not be graded by the retinal specialist because of insufficient image quality. Of the 2063 images with sufficient quality, 1823 (88.4%) showed no or mild DR, and 240 (11.6%) showed referable DR, as graded by the retinal specialist. The DLA classified 577 (25.2%) images as ungradable due to insufficient quality. Of the 1674 images with sufficient quality for both the reference standard and the DLA, 1481 (88.5%) showed no or mild DR, and 193 (11.5%) showed referable DR, as graded by the DLA (Table 2).

The performance of the DLA in detecting referable DR is shown in Table 3. For all gradable images, the DLA achieved an AUC of 0.942 (95% CI, 0.920-0.964), a sensitivity of 85.1% (95% CI, 83.4%-86.8%), and a specificity of 95.6% (95% CI, 94.6%-96.6%). The DLA showed consistent performance across different subgroups of patients stratified by history of HBP, sex, BMI, age, HbA1c, and diabetes duration. However, it showed superior performance in the subgroups with type 1 diabetes (AUC 0.996 and 95% CI, 0.988-1.000 for type 1 diabetes vs 0.938, 95% CI, 0.915-0.962 for type 2 diabetes), UACR ≥ 30 mg/g (AUC 0.945, 95% CI, 0.944-0.946 for UACR ≥ 30 mg/g vs 0.931, 95% CI, 0.930-0.932 for UACR < 30 mg/g), or eGFR < 90 mL/min/1.73m² (AUC 0.971, 95% CI, 0.970-0.972 for eGFR < 90 mL/min/1.73m² vs 0.941, 95% CI, 0.940-0.942 for eGFR ≥ 90 mL/min/1.73m²) (Table 3, Figure S1).

Of the 2286 images, 389 images could not be graded by the DLA because of their insufficient quality but could be graded by the retinal specialist. A review of those 389 images indicated that most of them ($n = 364$ [93.6%]) had grayish-green gradual translucent ring artifacts, which may be caused by light leakage due to improper distance between the eyes and the camera. Some images ($n = 66$ [17.0%]) had glare artifacts, which could also be seen in some images with ring artifacts. A few images had the defect of improper exposure ($n = 3$ [0.8%]) or poor focus/optical path occlusion (eyelashes, eyelids, etc.) ($n = 19$ [4.9%]). There were 35 images that could be graded by the DLA but not by the retinal specialist. The most common features of these images were ring artifacts ($n = 11$ [31.4%]) and improper exposure ($n = 23$ [65.7%]). Of the 2286 images, 188 images (8.4%) could not be graded by the DLA or the retinal specialist. More than 80% ($n = 156$ [83.0%]) of those 188 images had ring artifacts (Table 4). Examples of typical images with insufficient quality can be found in Figure S2.

TABLE 2 Comparison of DLA and retinal specialist grading

	Retinal specialist grade						Total
	No DR	Mild NPDR	Moderate NPDR	Severe NPDR	PDR	Ungradable	
DLA grade							
No DR	1278	100	7	3	0	27	1415
Mild NPDR	36	45	10	2	0	0	93
Moderate NPDR	22	40	74	20	1	2	159
Severe NPDR	0	0	1	2	1	0	4
PDR	3	2	6	11	10	6	38
Ungradable	251	46	45	26	21	188	577
All	1590	233	143	64	33	223	2286

Abbreviations: DLA, deep learning algorithm; DR, diabetic retinopathy; NPDR, nonproliferative DR; PDR, proliferative DR.

TABLE 3 Area under the receiver operating curve (AUC), sensitivity, and specificity of the DLA in detecting referable DR with reference to a retinal specialist's grading

	No. of eyes	Referable diabetic retinopathy		
		AUC	Sensitivity, % (95% CI)	Specificity, % (95% CI)
ALL	1674	0.942 (0.920-0.964)	85.1 (83.4-86.8)	95.6 (94.6-96.6)
Type of diabetes				
Type 1 diabetes	53	0.996 (0.988-1.000)	100.0 (100.0-100.0)	97.7 (93.7-100.0)
Type 2 diabetes	1621	0.938 (0.915-0.962)	84.2 (82.4-85.9)	95.0 (94.5-96.6)
HBP history				
No history of HBP	1114	0.940 (0.910-0.970)	84.4 (82.3-86.6)	96.5 (95.4-97.6)
HBP	560	0.943 (0.911-0.975)	86.2 (83.4-89.1)	93.8 (91.8-95.8)
Sex				
Female	503	0.924 (0.897-0.977)	84.2 (81.0-87.4)	96.1 (94.4-97.8)
Male	1171	0.948 (0.924-0.971)	85.5 (83.4-87.5)	95.4 (94.2-96.6)
BMI (kg/m ²)				
BMI <24	515	0.937 (0.901-0.973)	81.0 (77.6-84.4)	94.7 (92.8-96.7)
BMI ≥24	1159	0.944 (0.916-0.972)	87.8 (85.9-89.7)	96.0 (94.8-97.1)
Age (y)				
Age ≤ 40	515	0.949 (0.907-0.992)	80.0 (76.5-83.5)	96.9 (95.4-98.4)
40 < Age ≤ 60	896	0.943 (0.914-0.972)	90.1 (88.2-92.1)	94.7 (93.3-96.2)
Age > 60	263	0.930 (0.876-0.984)	78.1 (73.1-83.1)	96.1 (93.8-98.4)
HbA1c, % (mmol/mol)				
HbA1c < 7 (53)	526	0.967 (0.931-1.000)	82.9 (79.5-86.0)	97.8 (96.5-99.0)
7 (53) ≤ HbA1c < 9 (75)	574	0.939 (0.902-0.975)	87.3 (84.6-90.0)	95.3 (93.6-97.0)
HbA1c ≥ 9 (75)	539	0.934 (0.898-0.969)	85.2 (82.2-88.2)	93.4 (91.3-95.5)
Diabetes duration (y)				
Diabetes duration < 1	705	0.960 (0.923-0.997)	87.2 (84.7-89.6)	97.0 (95.7-98.3)
Diabetes duration ≥ 1	969	0.931 (0.904-0.959)	84.4 (82.1-86.7)	94.5 (93.1-96.0)
Diabetes duration ≥ 5	558	0.918 (0.885-0.952)	84.9 (81.9-87.9)	92.6 (90.4-94.8)
Diabetes duration ≥ 10	299	0.910 (0.867-0.953)	85.7 (81.7-89.7)	91.9 (88.9-95.0)
UACR (mg/g)				
UACR < 30	1129	0.931 (0.930-0.932)	82.5 (80.3-84.7)	95.8 (94.6-97.0)
UACR ≥ 30	356	0.945 (0.944-0.946)	85.7 (82.1-89.3)	94.7 (92.3-97.0)
eGFR (mL/min/1.73m ²)				
eGFR ≥ 90	1553	0.941 (0.940-0.942)	85.5 (83.7-87.2)	95.6 (94.5-96.6)
eGFR < 90	86	0.971 (0.970-0.972)	84.6 (77.0-92.2)	94.5 (89.7-99.3)

Abbreviations: AUC, area under the receiver operating curve; DLA, deep learning algorithm; DR, diabetic retinopathy; eGFR, estimated glomerular filtration rate; HbA1c, glycosylated hemoglobin; HBP, high blood pressure; UACR, urine albumin-to-creatinine ratio.

The DLA gave 67 false positives and 22 false negatives for referable DR. We then analyzed the reasons for inconsistency between the DLA and the retinal specialist, which are presented in Table 5. The most common reasons for false-positive classification were the misdiagnosis of retinal microaneurysm as intraretinal hemorrhage ($n = 25$ [37.3%]) and the misidentification of

arteriovenous crossing ($n = 19$ [28.4%]) as venous beading. The other reasons included retinal vessel occlusion ($n = 2$ [3.0%]), age-related macular degeneration ($n = 4$ [6.0%]), macular holes ($n = 1$ [1.5%]), congenital vascular malformation ($n = 1$ [1.5%]), and congenital optic papillary malformation ($n = 1$ [1.5%]), all of which were misclassified as referable DR. However, approximately 20.9%

**TABLE 4** Features of ungradable images

Features	Gradable by retinal specialist/ungradable by DLA (n = 389)	Gradable by DLA/ungradable by retinal specialist (n = 35)	Ungradable by retinal specialist and DLA (n = 188)
Ring artifact (improper distance between the eyes and the camera)	364	11	156
Glare artifact (reflection of optical lens)	66	4	83
Improper exposure)	3	23	21
Poor focus or occlusion of optical path	19	0	35

Abbreviations: DLA, deep learning algorithm.

TABLE 5 Features of false positives and false negatives in the identification of referable diabetic retinopathy by the DLA

Reason	No.	Proportion (%)
False positives	67	100
Retinal microaneurysm misdiagnosed as intraretinal hemorrhage	25	37.3
Arteriovenous cross signs mistaken for venous beads	19	28.4
Retinal vessel occlusion	2	3.0
AMD	4	6.0
Macular hole	1	1.5
Congenital vascular malformation	1	1.5
Congenital optic papillary malformation	1	1.5
Normal retinal microvessels misdiagnosed as IRMA	2	3.0
Normal fundus with glare and/or stain misdiagnosed as exudates	12	17.9
False negatives	22	100
Linear intraretinal hemorrhage mistaken for a blood vessel	13	59.1
Omission of IRMA	1	4.5
Omission of venous beading	1	4.5
Others with unknown reasons	7	31.8

Abbreviations: AMD, age-related macular degeneration; DLA, deep learning algorithm; IRMA, intraretinal microvascular abnormalities.

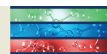
of false-positive images had no abnormal ocular findings, including some images in which normal retinal microvessels were misidentified as intraretinal microvascular abnormalities (IRMA) (n = 2 [3.0%]) and some images in which a glare and/or stains on a normal fundus were misidentified as exudates (n = 12 [17.9%]). An analysis of false-negative cases (n = 22) revealed that more than half of the images displayed linear intraretinal hemorrhage mistaken for blood vessels (n = 13 [59.1%]), and

seven images (31.8%) had been misclassified for unknown reasons. The remaining reasons included omission of IRMA (n = 1 [4.5%]) and venous beading (n = 1 [4.5%]). Examples of typical false-negative and false-positive images can be found in Figures S3 and S4.

4 | DISCUSSION

The prevalence of DR is increasing worldwide, and the reported prevalence ranged from 18.45% to 28.8% among DM patients in China.²²⁻²⁵ Our study demonstrated that the prevalence of any DR in patients with diabetes was 27.9%. DR does not usually have any obvious symptoms until it progresses to vision loss. Therefore, the annual screening for DR as well as other chronic complications related to diabetes is crucial in routine diabetes care. Early identification, assessment, and treatment are helping to reduce the overall burden of vision loss. Our DLA system is used in parallel with the clinical workflow, and it takes only 2 minutes from the beginning of retinal fundus image acquisition to the output of the results, indicating that it could rapidly screen a large number of patients and free the clinician from repetitive work.

Recently, a growing number of studies have been published on the accuracy of DLA for DR screening. Gulshan et al validated their DLA using approximately 10 000 retinal images retrieved from two publicly available databases (EyePAC-1 and Messidor-2) and achieved excellent performance (AUC 0.99; sensitivity and specificity >90%) for referable DR.¹¹ Li et al tested their DLA using 13 657 images from independent, multiethnic datasets and achieved an AUC, sensitivity, and specificity of 0.955, 92.5%, and 98.5%, respectively, for referable DR.¹⁵ Although these studies provided excellent insight, their DLA were validated against public databases comprising mainly high-quality photographs. In real-world circumstances, the quality of the photographs will be lower, which may reduce the performance of DLA.



However, a prospective study conducted in two tertiary eye care centers in South India showed that their DLA performed excellently in the detection of referable DR, with 88.9% to 92.1% sensitivity, 92.2% to 95.2% specificity, and an AUC of 0.963 to 0.980,¹⁶ which was comparable to the results of previous studies using public databases. Our DLA, which also used real-world screening data and nonmydriatic fundus photographs, showed similar performance (AUC 0.942, sensitivity 85.1%, and specificity 95.6%) to this prospective study conducted in South India. Therefore, our DLA offers a user-friendly, efficient, and professional tool for the early detection of referable DR in diabetic centers. In addition, we expect it to be beneficial in low-resource areas where specialized ophthalmologists are not available.

Our study showed that the DLA had better performance in the subgroups of patients with type 1 diabetes, albuminuria, and an eGFR < 90 mL/min/1.73m². It was reported that compared with patients with type 2 diabetes, those patients with type 1 diabetes were more likely to develop DR^{18,26} and that chronic kidney disease (CKD), low eGFR, and/or high UACR was associated with DR.^{18,27} Our research also found that the reduced eGFR levels and increased UACR levels were associated with the severity of DR. The performance of our DLA among the type 1 diabetes, albuminuria, and lower eGFR groups suggested that it performed better in patients with a high risk of DR. We speculate that patients with type 1 diabetes, albuminuria, and lower eGFR may have more typical DR lesions. However, the sample size of type 1 diabetes was small, and a large sample size is needed for further verification.

Older subjects are prone to media opacities,²⁸ and the clinical features of hypertensive retinopathy are similar to those of DR. Therefore, age and hypertension are sources of potential errors and may affect the performance of DLA. However, our DLA performed consistently in the subgroups of different ages and subgroups with or without a history of HBP, as well as in the subgroups stratified by sex, BMI, HbA1c, and diabetes duration. Therefore, our DLA performed excellently and consistently in those different subgroups, while it performed better in the subgroups with a high risk of DR (eg, type 1 diabetes and kidney impairment).

Poor-quality images are inevitable during the image acquisition process. In real-world screenings, the rates of poor-quality or ungradable images have been reported to be as high as 20%.^{28,29} Sufficient image quality is a key prerequisite for a reliable automatic DR detection system.³⁰ Here we analyzed the reasons for the images that cannot be graded by DLA. In our study, the DLA classified 25.2% as ungradable images due to insufficient quality; the most common reason for poor image quality was

the existence of ring artifacts (90%), which may be caused by light leaks due to improper distance between the eyes and the camera. The fundus photograph quality control model embedded in the image acquisition program can provide an automatic classifier for image quality, with the ability to effectively recognize low-quality images in real time and prompt image recapture, thereby reducing unqualified images.

Herein, we also explored the characteristics of misclassifications (false negatives and false positives) that occurred when our DLA was used to detect DR; this analysis will help us to better understand the functional weaknesses of our DLA and identify strategies to reduce errors in the future. Nearly 80% of the false-positive images displayed abnormal retinal features, which may have benefited from a referral when sent to the ophthalmologist. Among the false-positive images, 37.3% involved retinal microaneurysm misdiagnosed as intraretinal hemorrhage, and 28.4% involved arteriovenous crossing misidentified as venous beading. The sensitivity of our DLA grading for the referable DR was 85.1%, which would result in many missed cases. We analyzed the reasons for false-negative classification and found that nearly 60% of false-negative cases were shown to involve undetected linear intraretinal hemorrhage. Optimization of our DLA through further training on the above lesions may help to improve its fine-grained classification capabilities in the future.

While the results were encouraging, our study also had a few limitations. First, all retinal fundus images were from one hospital. Different equipment settings and camera systems would impact DR screening images and thus affect the performance of the DLA. To further validate our model, we need to collect more retinal fundus images from different hospitals in future research; we plan to apply this system in the community hospitals of our hospital alliance. Second, our DLA can only predict the DR grade but cannot identify diabetic macular edema, another major cause of vision loss in patients with diabetes. Nondiabetic retinopathy also cannot be identified because our system is trained only for DR. Third, the reference standard used for this study was based on one retinal specialist. However, he is an experienced ophthalmologist with more than 12 years of experience.

In conclusion, our DLA-based DR grading software showed excellent and consistent performance in the detection of referable DR in retinal fundus images across a majority of subgroups in a real-world diabetic center and showed superior performance in patients with type 1 diabetes and diabetic kidney disease. Moreover, we expect that it will prove to be highly useful when applied in primary care settings where qualified eye care professionals are not always available; further application and



research is needed to improve the clinical validity of this algorithm.

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CONFLICT OF INTEREST

No potential conflicts of interest relevant to this article were reported.

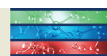
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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

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